

Project Preparation Report

Jacob Josiah Webber

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1 Introduction

BEEP BOOP IM A CITER This report details the preparation for a final project for an MSc in High Performance Computing. This project aims improve upon current techniques used to run 3 dimensional finite-difference time-domain (FDTD) simulations on the GPU. The problem with existing methods that do this is that a bounding box is used to store all points in the simulation. This is inefficient in its use of memory. This project hopes to devise and implement methods that make better use of limited GPU high bandwidth memory.

This report opens with a critical review of two previous dissertations that cover different topics to this project. There will then be a brief background and literature review, which will describe some of the theory behind room acoustics, some alternatives to the FDTD technique and an overview of the state of the art in this field. The report will then provide a comprehensive plan of action for how the work will be undertaken, including a finalised project proposal, and a work plan and risk analysis. This report will then detail any initial findings made during the project preparation stage, before concluding.

2 Previous Dissertation Review

This section contains a review of two previous dissertations submitted by other students. The reports selected for review are not chosen to be particularly relevant in their technical content to the topic of this project, but are useful as examples of high quality work at the level required.

2.1 Dissertation 1 - “Fulfilling Solvency II regulations using GPUs” by Ludovic Capelli, 2016

This first dissertation discusses work done to speed up the computation of financial solvency for insurance companies, in line with new EU wide regulations. These regulations involve Monte-Carlo simulations that require a very large amount of computational resource to comply with. The project improved upon previously state of the art performance on the GPU by a factor of two and was awarded a distinction level mark.

The topic seems to be a particularly suitable one for a project in High Performance Computing. Present technology in use by industry would take approximately a year to calculate whether an insurance provider is solvent. By this stage the data fed in will be out of date and the impact of any insolvency may have already been felt. It is therefore necessary and appropriate to use higher performance techniques to do these calculations, even if doing so requires the additional programming and hardware costs associated with such methods.

The dissertation outlines in an appropriate amount of detail the background to the regulations and details of insurance insolvency, striking a balance between not being a dissertation on finance but providing adequate background for the reader to understand the nature of and motivation behind the calculations performed. This content is included briefly in the introduction and in more detail in the Literature review section.

A comprehensive description of GPU programming techniques is provided for the reader. This includes a generous amount of well styled diagrams describing the GPGPU programming paradigm in both CUDA and OpenMP 4.0. Particular attention is given to achieving optimal memory usage. This information will be useful for undertaking a project on 3d room acoustics modelling - where memory accesses are the major bottleneck.

There are few criticisms that can be offered of what is a very well written dissertation covering a project that was measurably successful in its outcomes. However, it could be said that some of the discussion of computing basics is a little unnecessary for a dissertation of this level, in particular the description of a CPU as the “brain” of the computer.

This dissertation is generally well formatted and easy to read, using the standard EPCC template in L^AT_EX. However, there is perhaps a little too much included in the preface, with a full table of contents, table of figures and index of tables, along with blank pages in between these. This means that in a document of 69 pages, the actual report does not start until the 14th page.

To conclude the review of this report, it is clearly a very strong piece of work. Good results were achieved with a 100% speed up on previous work. While Mr Capelli’s dissertation was not selected for review on the basis of its relevance to the this project, the detailed discussion of GPGPU programming and memory management will be extremely useful background for undertaking future work, even when applied to a completely different problem.

2.2 Dissertation 2

3 Background and Literature Review

Acoustics is the study of the physical aspects of sound. Sound is our perception of varying pressure with time in the air around us. The way that this sound, or pressure variation propagates through the air around us can be modelled using a well know differential equation known as the wave equation. In one dimension this is

$$\frac{\partial^2 \mathbf{p}}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2 \mathbf{p}}{\partial t^2} = 0 \tag{1}$$

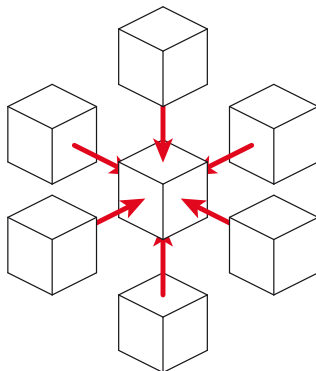


Figure 1: Diagram showing 3 dimensional Von Neumann stencil. Credit: Wikimedia Commons.

where $\mathbf{p}$ is acoustic pressure, and c is the speed of sound. This result is derived by, among others, Richard Feynman in [1]. It is trivial to generalise this for 3 dimensional systems, which yields

$$\nabla^2 \mathbf{p} - \frac{1}{c^2} \frac{\partial^2 \mathbf{p}}{\partial t^2} = 0 \quad (2)$$

It should be noted that in both of these equations $\mathbf{p}$ is the deviation from the ambient air pressure.

Much of the work in the field of acoustics entails finding solutions to this equation in various mediums and bounding conditions. In the case of room acoustics, the medium of vibration is air, so c corresponds to the speed of sound in air. The nature of air is common to all spaces, with very minor changes depending on altitude or humidity, the majority of room specific acoustic features are determined by the boundary conditions. For large rooms with complex boundaries it is not possible to solve this equation analytically, so numerical solutions are found using the FDTD technique.

Finite difference methods are a way of approximating and discretising differential equations so that they can be solved numerically. Their use effectively turns the problem of solving a differential equation into a large stencilling operation. A stencilling operation, as shown in figure 1, is the process of replacing each element in a virtual 3d space with the sum of its neighbours.

A simple equivalence between a continuous derivative and a finite difference in one dimensional space is:

$$\frac{df(x)}{dx} \approx \frac{f(x+h) - f(x-h)}{2h} \quad (3)$$

From this second order finite difference schemes can be derived.

Stencilling operations are in their nature memory bound problems as they involved a lot of data accesses per, relatively simple, calculation. This means that GPUs, with their high bandwidth memory, have been shown to be ideally suited to perform these operations [2]. However, while GPUs have the best performance, they also have limited amounts of memory. This provides

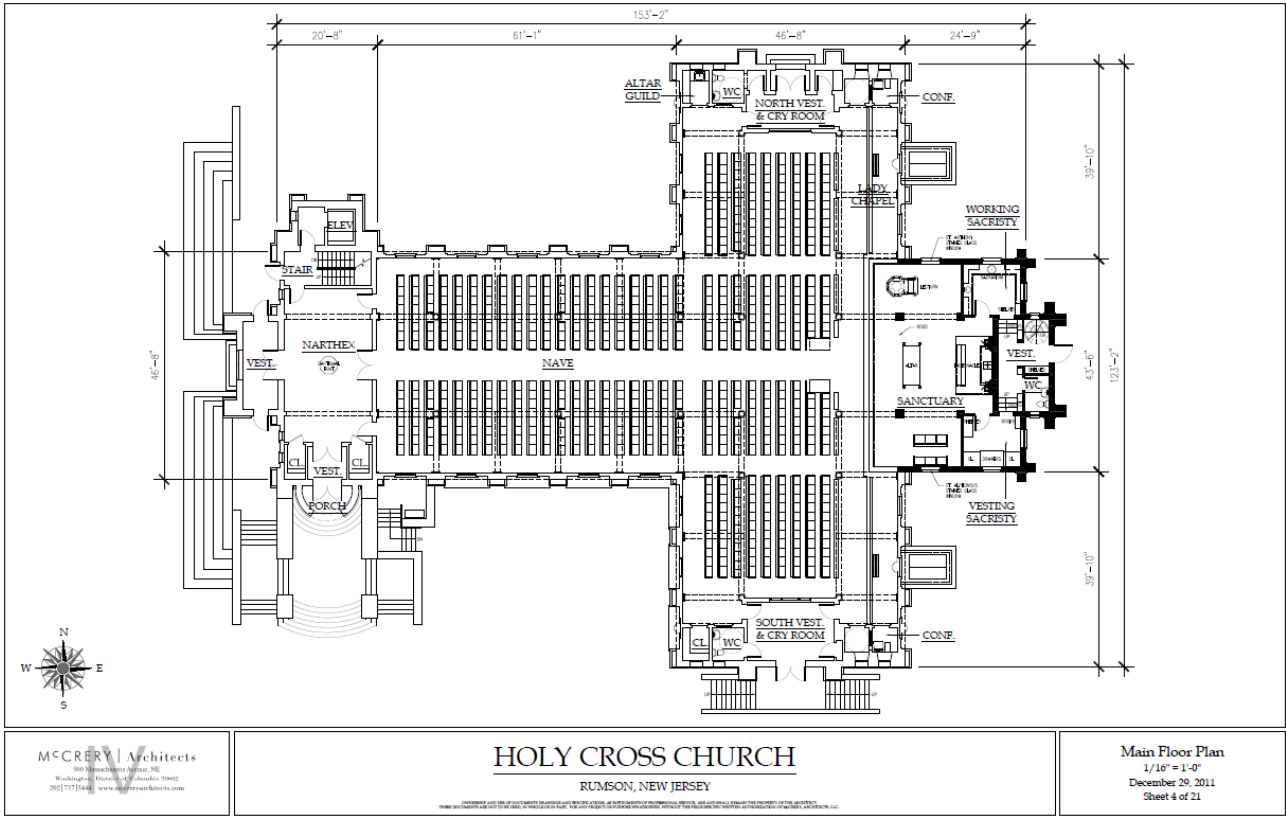


Figure 2: A diagram showing a schematic of a church building. Credit: McCrery Architects.

problems when trying to do acoustic models as such models require a lot of data points to model even moderately sized rooms. The wavelength of sound at 20 kHz, the highest frequency audible by humans, is 17 mm. To avoid aliasing or other artefacts, as well as to maintain numerical stability, it is necessary to use multiple points per wavelength.

There is a great deal of literature on the number of points required to be stored for a given simulation, the detail of which goes outside the scope of this report. [3] shows 10s of points per wavelength being used, whereas in previous work by the Acoustics and Audio Group at the University of Edinburgh (AAG) a ratio of $\sqrt{3}$ between time step and sample length has been shown to at least maintain numerical stability. This ratio is known as the Courant limit and a full derivation is shown in [4]. In either case the memory requirements are large. [5] shows a space of 4775 m² as requiring 24 GB of memory to simulate. In contrast, The Royal Albert hall is popularly claimed to have a volume of approximately 80 000 m².

Another problem with the spacial expense of these computations is caused by the nature of three dimensional arrays. These are cuboid in shape. While this has the benefit of allowing a point's memory address to be easily derived from its spacial location, it does lead to memory being wasted when rooms of less regular shape are being simulated. As an example, figure 2 shows a very common building type: the church. With this shape a large proportion of the space allocated in memory is assigned to space outside the simulation. Without careful programming this empty space could also waste compute, were the stencilling operation to be unnecessarily applied to this region.

Three dimensional arrays are stored in computers as one dimensional arrays. The elements in one dimension concatenated to form a planar representation. These representations are then concatenated again to form a longer array representing the 3d space. This means that when

the stencilling operation is performed many of the neighbouring elements in space are far apart in memory. Non-contiguous device main memory access is much slower than contiguous access. In addition to this, this non-contiguous access increases the number of cache misses as cache lines are loaded in contiguous blocks.

4 Proposal

The purpose of this project is to try to improve memory usage for irregularly shaped rooms and memory locality in large 3d FDTD acoustics simulations.

Use an array of structs instead!

- Store enough points for CUDA kernel to operate on.
- Only store structs with at least one point inside the room.
- Experiment with different sized arrays within the struct to see if caching benefits can be yielded.

5 Work Plan

I will write the code

6 Risk Analysis

GPUs are dangerous

7 Initial Findings

I'm basically sure it works.

Listing 1: First 2d example

```
struct linked_struct {
    int up;
    int down;
    int left;
    int right;
    float value[N][M];
    float old_value[N][M];
    char inside[N][M];
};
```

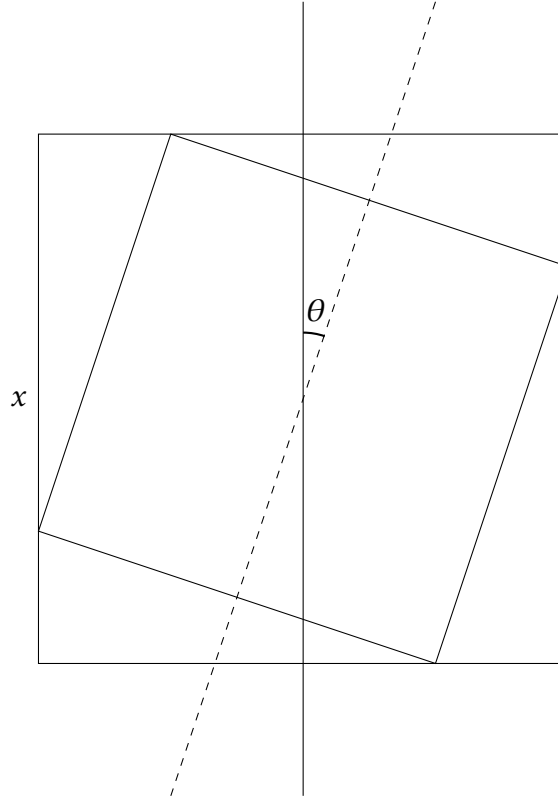


Figure 3: Diagram showing unit square rotated by θ within bounding square, where x is the side of the bounding square.

8 Conclusion

References

- [1] Richard Feynman, 1969, *Lectures in Physics, Volume 1*, Addison Publishing Company, Addison
- [2] Andreas Schfer, Dietmar Fey, *High Performance Stencil Code Algorithms for GPGPUs*, Procedia Computer Science, Volume 4, 2011, Pages 2027-2036, ISSN 1877-0509, <http://dx.doi.org/10.1016/j.procs.2011.04.221>. (Accessed: 5 April 2017)
- [3] D. W. Zingy , H. Lomax , and H. Jurgens , *High-accuracy finite-difference schemes for linear wave propagation*, SIAM J. Sci. Comput. 17, 328346, 1996.
- [4] Hamilton, B. (2016) *Finite Difference and Finite Volume Methods for Wave-based Modelling of Room Acoustics*. PhD thesis. University of Edinburgh. Available at: <http://www2.ph.ed.ac.uk/~s1164563/thesis/hamilton2016phdthesis.pdf> (Accessed: 5 April 2017)
- [5] Webb, C.J. (2009) *Parallel computation techniques for virtual acoustics and physical modelling synthesis*. PhD thesis. University of Edinburgh. Available at: http://www2.ph.ed.ac.uk/~cwebb2/Site/Docs/CJWebb_thesis_FINAL.pdf (Accessed: 5 April 2017)